数组计数法

人员

臧晨云 隋溟琛 霍佳铭 于昊华 刘奕男 刘嘉浩 刘子郴 李景翾 何佩瑜 林钰明 初星熠 张高 刘克阳 到课

上周作业检查

上周作业链接: https://cppoj.kids123code.com/contest/3191

王向东老师周六一点半C++数组元素移动

刷新

#	用户名	姓名	编程分	时间	A	B	C	D	E	F	G	H	I	J	K	L
1	zangchenyun	臧晨云	1200	215	100	100	100	100	100	100	100	100	100	100	100	100
2	yuhaohua	于昊华	1200	216	100	100	100	100	100	100	100	100	100	100	100	100
3	chuxingyi	初星熠	1200	223	100	100	100	100	100	100	100	100	100	100	100	100
4	suihaochen	隋溟琛	1200	236	100	100	100	100	100	100	100	100	100	100	100	100
5	zhanggao	张高	1200	239	100	100	100	100	100	100	100	100	100	100	100	100
6	huojiaming	霍佳铭	1200	301	100	100	100	100	100	100	100	100	100	100	100	100
7	lijingxuan	李景翾	1100	162	100	100	100	100	100	100	100	100	100		100	100
8	liuzichen123	刘子郴	900	173	100	100	100	100	100	100	100	100	100	0		
9	hepeiyu	何佩瑜	900	196	100	100	100	100	100	100	100	100	100			
10	chenzhiheng	陈之恒	900	257	100	100	100	100	100	100	100	100	100			
11	liuyinan	刘奕男	800	121	100	100	100	100	100	100		100	100			
12	zhaoyuxuan	赵裕轩	800	123		100	100	100	100	100	100		100		100	
13	liukeyang	刘克阳	800	165	100	100	100	100	100	100	100	100				
14	liujiahao	刘嘉浩	700	100	100	100	100	100	100	100	100					
15	linyuming	林钰明	400	0		100	100	100	100							

作业

https://cppoj.kids123code.com/contest/3297 (课上讲了 A ~ F 题, 作业是 F 题必做, G H 选做)

课堂表现

今天上课主要给同学们讲了数组计数法, 关键需要同学们深刻理解 cnt 数组的含义与使用, 建议同学们课后好好复习复习课上的几道题目。

课堂内容

换位置

```
#include <bits/stdc++.h>

using namespace std;

int a[105];

int main()
{
    int n;
```

```
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    int maxx = -10000000, minn = 100000000;
    for (int i = 1; i <= n; i++) {
        if (a[i] < minn) {
            minn = a[i];
        }
        if (a[i] > maxx) {
            maxx = a[i];
        }
    }

    int x, y;
    for (int i = 1; i <= n; i++) {
        if (a[i] == minn) {
            x = i;
        }
        if (a[i] == maxx) {
            y = i;
        }
    }

    swap(a[x], a[y]);
    for (int i = 1; i <= n; i++) {
        cout << a[i] << " ";
    }
    return 0;
}
```

移动数组元素

```
#include <bits/stdc++.h>

using namespace std;

int a[105];

int main()
{
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    int maxx = -10000000, minn = 100000000;
    for (int i = 1; i <= n; i++) {
        if (a[i] < minn) {
            minn = a[i];
        }
```

```
        if (a[i] > maxx) {
            maxx = a[i];
        }
    }

    int x, y;
    for (int i = 1; i <= n; i++) {
        if (a[i] == minn) {
            x = i;
        }
        if (a[i] == maxx) {
            y = i;
        }
    }

    swap(a[x], a[1]);
    swap(a[y], a[n]);
    for (int i = 1; i <= n; i++) {
        cout << a[i] << " ";
    }
    return 0;
}
```

求n个数中每个数出现的次数

```
#include <bits/stdc++.h>

using namespace std;

int a[105], cnt[15];

int main()
{
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    for (int i = 1; i <= n; i++) {
        cnt[a[i]]++;
    }
    for (int i = 1; i <= 10; i++) {
        if (cnt[i] != 0) {
            cout << i << " " << cnt[i] << endl;
        }
    }
    return 0;
}
```

求n个数中出现次数最多的数

```
#include <bits/stdc++.h>

using namespace std;

int a[105], cnt[15];

int main()
{
    int n;
    cin >> n;
    for (int i = 1; i <= n; i++) {
        cin >> a[i];
    }
    for (int i = 1; i <= n; i++) {
        cnt[a[i]]++;
    }

    int maxx = -100000;
    for (int i = 1; i <= 10; i++) {
        if (cnt[i] > maxx) {
            maxx = cnt[i];
        }
    }
    int x;
    for (int i = 1; i <= 10; i++) {
        if (cnt[i] == maxx) {
            x = i;
        }
    }

    cout << x << endl;
    return 0;
}
```

声音识别

```
#include<bits/stdc++.h>
using namespace std;
long long a[100009];
int cnt[105];
int main(){
    long long c,d,f,e,g=0,n;
    cin>>n;
    for(int i=1;i<=n;i++){
        cin>>a[i];
        cnt[a[i]]++;
    }
    for(int i=1;i<=100;i++){
        if(cnt[i]!=0){
            g++;
```

```
        }
    }
    cout<<g<<endl;
    for(int i=1;i<=100;i++){
        if(cnt[i]!=0){
            cout<<i<<" "<<cnt[i]<<endl;
        }
    }
    return 0;
}
```

[NOIP 2005 普及组] 校门外的树

```
#include <bits/stdc++.h>

using namespace std;

int w[10000+5];

int main()
{
  int n, m; cin >> n >> m;
  for (int i = 0; i <= n; i++) {
    w[i] = 1;
  }
  for (int i = 1; i <= m; i++) {
    int l, r; cin >> l >> r;
    for (int j = l; j <= r; j++) {
      w[j] = 0;
    }
  }
  int sum = 0;
  for (int i = 0; i <= n; i++) {
    if (w[i] == 1) {
      sum ++;
    }
  }
  cout << sum << endl;
  return 0;
}
```